Politécnico do Porto
Instituto Superior de Engenharia do Porto

Development of an Integrated System for Acoustic Calibration of Stethoscopes

Rodrigo Pires

LEEC
Degree in Electrical and Computer Engineering

ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

Departamento de Engenharia Eletrotécnica
Instituto Superior de Engenharia do Porto

April, 2026

This report partially satisfies the requirements defined for the Project/Internship course, in the 3rd year, of the Degree in Electrical and Computer Engineering.

Candidate: Rodrigo Pires, No. 1210697, 1210697@isep.ipp.pt

Scientific Guidance: Luis Pinto Coelho, lfc@isep.ipp.pt

Departamento de Engenharia Eletrotécnica
Instituto Superior de Engenharia do Porto
Rua Dr. António Bernardino de Almeida, 431, 4200–072 Porto

April, 2026

Abstract

Aqui deverá ser apresentado o resumo de todo o trabalho efetuado. Esta secção não deverá exceder uma página.

Deve contextualizar o problema que pretende resolver ou a hipótese que irá formular, procure evidenciar as vantagens e desvantagens (se as houver) da solução encontrada, como também a forma através da qual a solução/hipótese foi validada. Neste último ponto, deverá referir-se aos desenvolvimentos efetuados, e à forma como validou (conformidade) e avaliou (desempenho) a solução encontrada.

O documento deve conter sempre duas versões do resumo: uma primeira no idioma do texto principal e a segunda num outro idioma. Este *template* assume que os dois idiomas em consideração são sempre Português e Inglês, assim, a classe irá colocar os cabeçalhos respetivos de acordo com o idioma selecionado nas opções da classe no ficheiro `main.tex`.

Keywords: Lista, separada por vírgulas, de palavras, frases, ou acrónimos chave no âmbito do trabalho descrito neste texto.

Resumo

The summary of all the developed work should be presented here. This section should not exceed one page.

Start the abstract with the contextualization of the problem you intend to solve or the hypothesis you will formulate. Try to highlight the advantages and disadvantages (if any) of the solution found, as well as the way in which the solution/hypothesis was validated. In this last point, you should refer to the developments made, and to the way you validated (compliance) and evaluated (performance) the solution found.

The document must always contain two versions of the abstract: a first in the language of the main text and the second one in another language. This template assumes that the two languages are always Portuguese and English, therefore, the class will place the correct section headers according to the language selected in the class options in the `main.tex` file.

Palavras-Chave: Comma separated list of words, phrases, or key acronyms within the scope of your developed work.

Contents

List of Figures **vii**

List of Tables **ix**

Listings **xi**

List of Algorithms **xiii**

List of Symbols **xv**

1 Introduction **1**

1.1 State of the Art/Contextualization 1

1.2 Project Description 2

1.2.1 Objectives 2

2 Methodology **3**

2.1 Literature Review 3

2.1.1 Sound Generation Technologies 3

Power Amplifier 4

2.1.2 Sound Acquisition Technologies 5

Pre-Amplifier 6

2.1.3 Signal Digitization: Analog-to-Digital Conversion (ADC) . . 8

References **9**

List of Figures

List of Tables

2.1 Conversion of Microphone Sensitivity (dBV/Pa) to Raw Voltage Output (mV/Pa) 6
2.2 Datasheet Specification Comparison: NE5532P vs. TL072 7

Listings

List of Algorithms

List of Symbols

Symbol	Description	Units
f	força	N
i	corrente	A
M	massa	kg
P	potência	W
x	deslocamento	m
θ	deslocamento angular	rad
ω	velocidade angular	$\mathrm{rad\,s^{-1}}$

Chapter 1

Introduction

1.1 State of the Art/Contextualization

Stethoscopes remain the most fundamental and prevasive diagnostic tools in clinical practice for the auscultation of cardiopulmonary and vascular sounds. Currently, traditional acoustic stethoscopes dominate over 90% of the global market due to their high durability, low cost, and independence from electrical power sources [1]. However, purely acoustic auscultation is inherently limited not only by the fallibility of human auditory perception but also by the unrecognized physical degradation of the instrument over time. Studies have demonstrated that general practitioners miss up to 57% of significant valvular heart diseases when relying on standard stethoscopes [2].

While frequently attributed to human auditory limitations, this high error rate may be significantly composed by unnoticed material failures; such as compromised acoustic seals in the eartips, hardened tubing, or worn diaphragms. These physical degradations silently alter the stethoscope's frequency response, attenuating vital low-frequency biological sounds before they ever reach the clinician's ear. To mitigate these human diagnostic errors, the medical field is experiencing the gradual emergence of electronic stethoscopes designed to amplify, filter, and digitize biological sounds.

This technological shift, while clinically beneficial, introduces a critical new engineering problem: these electronic devices actively alter and process the raw audio

signal. Because both the acoustic integrity of traditional tubing and the digital processing of modern devices directly impact diagnostic accuracy, the development of a precise calibration system to verify signal fidelity has never been more urgent.

1.2 Project Description

This project proposes the development of an acoustic calibration system designed to characterize the frequency response, and signal fidelity of stethoscopes. The proposed architecture consists of two primary domains: a physical hardware module for sound generation and detection, and a software platform for control, data acquisition, and spectral analysis.

1.2.1 Objectives

By the end of the project, the developed system should be able to:

- **Simulate Cardiopulmonary Acoustics:** Generate controlled acoustic signals that mimic the frequency spectrum of biological sounds, particularly in the frequency range for cardiopulmonary auscultation.
- **Capture High-Fidelity Audio Data:** Accurately measure the stethoscope's output signal using a high-fidelity microphone and data acquisition system, ensuring that the measurement setup itself does not introduce significant distortions.
- **Analyze Frequency Response:** Analyze the acquired audio data to compute the stethoscope's frequency response curve utilizing python-based spectral analysis techniques.
- **Establish a Repeatable Protocol:** Develop a standardized testing protocol that can be easily replicated for routine stethoscope calibration in clinical settings, ensuring consistent and reliable performance assessments.

Chapter 2

Methodology

2.1 Literature Review

First and foremost, for this acoustic calibration of stethoscopes it was needed to, understand in which frequencies the cardiopulmonary sounds operate in. This is important to know which frequencies we need to be able to reproduce in our calibration system, and also measure the stethoscope's response.

According to studies, the frequency range of heart sounds is typically between 5 Hz and 600 Hz, while lung sounds can range from 100 Hz to 2000 Hz [4]. Other studies have also shown that the most important frequencies for heart sounds and diseases are typically between 20 Hz and 150 Hz, while for lung sounds, the most important frequencies are between 100 Hz and 1000 Hz [5]. This means that our calibration system should prioritize the ability to reproduce sounds in these frequency ranges, as they are the most relevant for clinical auscultation.

2.1.1 Sound Generation Technologies

Traditional audio testing frequently relies on standard dynamic speakers. However, dynamic speakers operate by displacing free air.

Because a medical stethoscope is clinically designed to capture vibrations through high-impedance solid mediums (such as human tissue and fluid), utilizing an air-coupled speaker introduces significant energy loss, particularly in the low-frequency cardiovascular bands (20 Hz - 150 Hz).

To overcome this limitation, modern biomedical testing methodologies require the use of surface transducers or exciters. Instead of pushing air, surface transducers are designed to directly vibrate the solid object to which they are attached. When a surface transducer is mechanically coupled with a tissue-simulating medium (such as silicone or acoustic gelatine), it can accurately replicate the acoustic impedance of a human chest [6].

Following the justification for structure-borne sound converters, three commercially available surface transducers were selected for evaluation in this project:

- **Dayton Audio TT25-8 PUCK:** This is a compact, high-performance tactile transducer designed for low-frequency applications. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 19€.
- **Dayton Audio HDN-8 Weatherproof:** This is a high-performance, weatherproof tactile transducer designed for outdoor use. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 25€.
- **Dayton Audio DAEX25FHE-4:** This is a compact and affordable exciter with a frequency response of 70 Hz to 20 kHz and comes at a price range of around 15€.

For the sake of this project, the Dayton Audio DAEX25FHE-4 exciter was selected as the most suitable option for our stethoscope calibration system. This choice was based on its ability to produce strong vibrations in the relevant frequency range for cardiopulmonary sounds, as well as its affordability and compact size, which are important considerations for a practical calibration system.

Power Amplifier

To drive the Dayton Audio DAEX25FHE-4 exciter, the reference signal generated by the central control software must be passed through a power amplifier. The primary requirement for this component is signal linearity; the amplifier must deliver a perfectly flat frequency response from 20 Hz to 2000 Hz without introducing Total Harmonic Distortion (THD) or low-frequency roll-off.

Two primary amplifier topologies were evaluated for the physical test rig:

- **Class-D Amplifiers (e.g., PAM8403 / TPA3116):** Class-D modules operate utilizing PWM. While extremely affordable and capable of operating on standard 5V USB power, the digital switching nature of Class-D topologies introduces high-frequency switching noise and poorer inherent linearity [7].

- **Class-AB Amplifiers (e.g., TDA2030A / LM1875):** Class-AB amplifier provides a continuous, highly linear analog output, ensuring a pure analog signal path across the entire target frequency spectrum. Literature demonstrates that Class-AB topologies consistently outperform Class-D stages in Total Harmonic Distortion (THD) metrics, frequently achieving distortion figures up to 30 dB better than their digital counterparts [8].

To ensure the acoustic reference signal remained free of high-frequency digital artifacts and low-frequency attenuation, a **Class-AB TDA2030A** audio amplifier module was selected. It comes at a highly cost-effective price point of around 2€.

2.1.2 Sound Acquisition Technologies

To accurately capture the acoustic output of the stethoscope, the microphone must satisfy two critical requirements: it must possess a physical form factor small enough to achieve an airtight seal with the stethoscope's eartip (approximately 5 mm in diameter), and it must feature a flat frequency response capable of registering extreme low-frequency cardiovascular sounds down to 20 Hz.

Two primary microphone technologies were evaluated for the acquisition stage:

- **MEMS (Micro-Electro-Mechanical Systems) Microphones:** Commonly utilized in modern mobile devices, MEMS microphones are exceptionally compact and digitally integrated. However, consumer-grade MEMS capsules are specifically engineered for human speech transmission. Consequently, they frequently incorporate built-in high-pass filters designed to attenuate wind and handling noise below 60 Hz. Utilizing a standard MEMS microphone would result in the severe attenuation of foundational S1, S2, and S3 heart sounds, rendering them unsuitable for this calibration system.

- **ECM (Electret Condenser Microphone) Capsules:** ECMs are traditional analog acoustic sensors utilized in audio measurement. A standard 6 mm ECM capsule provides the necessary compact form factor for an airtight eartip coupling. More importantly, unlike consumer MEMS devices, measurement-grade ECM capsules do not artificially filter low-frequency kinetic energy, providing a highly linear frequency response down to 20 Hz.

Given the low-frequency requirements of cardiopulmonary auscultation, a 6 mm ECM capsule was selected as the optimal sensor for the acquisition stage. Because the raw analog output of an ECM capsule is highly attenuated (typically in the millivolt range), this selection necessitates the integration of a low-noise pre-amplification stage prior to digital signal processing.

Table 2.1: Conversion of Microphone Sensitivity (dBV/Pa) to Raw Voltage Output (mV/Pa)

Microphone Example	**Sensitivity (dBV/Pa)**	**Voltage Output (mV/Pa)**
Generic Low-Cost ECM	-60 dB	1.00 mV
High-Sensitivity ECM	-40 dB	10.00 mV
Selected Prototype ECM	**-35 dB**	**17.78 mV**
Premium Measurement Mic	-20 dB	100.00 mV

As demonstrated in Table 2.1, microphone sensitivity operates on a logarithmic scale. A Generic Electret Condenser Microphone rated at -60 dB outputs approximately 1.00 millivolts per Pascal of acoustic pressure. By acquiring a higher sensitivity capsule rated at -35 dB, the electrical output increases drastically to 17.78 millivolts per Pascal, yielding a signal almost 18 times stronger for the exact same acoustic event.

This stronger signal reduces the gain requirements for the subsequent ADC pre-amplification stage, therefore minimizing the amplification of inherent electronic noise and improving the overall Signal-to-Noise Ratio (SNR) of the acquired data.

Pre-Amplifier

Despite selecting a high-sensitivity Electret Condenser Microphone (-35 dBV/Pa), the raw analog output generated by cardiopulmonary acoustics remains in the low millivolt range. To prevent quantization error and signal loss during analog-to-digital conversion, the microphone's output must be conditioned and boosted using a dedicated pre-amplifier stage. The selection of the Operational Amplifier (Op-Amp) for this stage is highly critical, as any inherent electronic noise introduced here will be amplified and permanently embedded into the acquired acoustic data.

Three pervasive operational amplifiers were evaluated for this signal conditioning stage:

- **LM358 (General Purpose Op-Amp):** While exceptionally affordable and common in basic microcontroller kits, the LM358 is a general-purpose operational amplifier not intended for high-fidelity audio applications. It suffers from significant crossover distortion and possesses a relatively high input noise profile, making it unsuitable for capturing delicate, low-amplitude respiratory frequencies.
- **TL072 (JFET-Input Op-Amp):** The TL072 is a widely utilized, audio-grade Op-Amp commonly found in electronic stethoscope designs. Its JFET inputs provide high input impedance, and it boasts a low Total Harmonic

Distortion (THD) profile alongside a low noise floor (typically 18 nV/$\sqrt{\text{Hz}}$). It is an excellent, low-cost option for general biological signal amplification.

- **NE5532P (Ultra-Low Noise Audio Op-Amp):** The NE5532 is historically considered the industry standard for professional audio pre-amplification. It features a high slew rate and an ultra-low noise floor (typically 5 nV/$\sqrt{\text{Hz}}$). While it draws slightly more quiescent current than the TL072, it offers superior linear performance and exceptionally clean gain across the entire 20 Hz to 20 kHz spectrum.

Table 2.2: Datasheet Specification Comparison: NE5532P vs. TL072

Parameter	**NE5532P**	**TL072**	**Relevance to Calibration Rig**
Input Architecture	BJT (Bipolar)	JFET	JFETs offer higher input impedance, but BJTs offer lower voltage noise.
Equivalent Input Noise	5 nV/$\sqrt{\text{Hz}}$	18 nV/$\sqrt{\text{Hz}}$	NE5532 is significantly quieter, crucial for capturing faint respiratory crackles without static.
Gain Bandwidth (GBW)	10 MHz	3 MHz	Both exceed the 20 kHz requirement, but NE5532 offers more linear gain at higher frequencies.
Slew Rate	9 V/μs	13 V/μs	Both are more than sufficient to accurately track rapid acoustic transients (e.g., S1/S2 heart valves snapping shut).

As detailed in Table 2.2, both operational amplifiers possess technical specifications that easily satisfy the baseline requirements for cardiopulmonary acoustic processing.

The primary challenge of this acquisition stage is the noise mitigation when amplifying the microvolt-level signals of faint respiratory anomalies. The NE5532P reduces the equivalent input noise down to 5 nV/$\sqrt{\text{Hz}}$, less than a third of the noise generated by the TL072. Given the necessity to capture low-amplitude anomalies—such as pulmonary crackles—without introducing electronic interference, the **NE5532P** was selected as the optimal Op-Amp for the pre-amplifier circuit. It comes at a highly cost-effective price point of around 1€.

2.1.3 Signal Digitization: Analog-to-Digital Conversion (ADC)

Following the analog pre-amplification stage, the signal must be digitized for computational analysis. The fidelity of the resulting Fast Fourier Transform (FFT) algorithms is heavily dependent on the resolution and sampling rate of the chosen Analog-to-Digital Converter (ADC).

Two primary digital conversion architectures were evaluated:

- **Microcontroller ADC (e.g., Arduino Uno 10-bit):**
- **Dedicated Audio ADC (16-bit USB Interface):**

References

[1] Transparency Market Research, "Stethoscope market - global industry analysis, size, share, growth, trends, and forecast, 2022-2031." Industry Report, 2022. [Online]. Available: https://www.transparencymarketresearch.com/. [Cited on page 1]

[2] S. K. M. Gardezi, S. G. Myerson, B. Prendergast, A. Prothero, A. Prendergast, M. Loudon, and J. D'Arcy, "Cardiac auscultation in diagnosing valvular heart disease: a comparison between general practitioners and cardiologists," *European Heart Journal*, vol. 38, no. suppl_1, 2017. [Cited on page 1]

[3] S. D. Poonguzhali, T. N. Prabakar, and V. Elamaran, "Fpga-based design and implementation of cascaded digital filters for enhanced auscultation in digital stethoscopes," *Circuits, Systems, and Signal Processing*, 2025.

[4] M. Tavakoli *et al.*, "Heart and lung sound measurement using an esophageal stethoscope with adaptive noise cancellation," *Sensors*, vol. 21, no. 20, p. 6757, 2021. [Cited on page 3]

[5] J. K. Roy, T. S. Roy, and S. C. Mukhopadhyay, "Heart sound: Detection and analytical approach towards diseases," in *Modern Sensing Technologies*, pp. 117–139, Springer, 2019. [Cited on page 3]

[6] M. Klum *et al.*, "Validation system for digital stethoscopes," *Current Directions in Biomedical Engineering*, vol. 5, no. 1, pp. 261–264, 2019. [Cited on page 4]

[7] Y. Zhang *et al.*, "The class d audio power amplifier: A review," *Electronics*, vol. 11, no. 19, p. 3244, 2022. [Cited on page 4]

[8] M. Duraij *et al.*, "A comparative study on class ab and class d amplifier topologies," in *2021 IEEE International Symposium on Power Line Communications and its Applications (ISPLC)*, pp. 1–6, IEEE, 2021. [Cited on page 5]